DATA SCIENCE & ARTIFICIAL INTELLIGENCE

Probability and Statistics

Statistics - 1

Lecture No. 09





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Recap of previous lecture



Topic

(1) Exponential Dist
(2) Uniform Dist.

Topics to be Covered



Topic

NORMAL DISTRIBUTION

HW8/P18 Traffic is moving at a certain junction with Rate 360 vehicles/hr then Find the prob that Headway b/n two successive vehicles lies b/n 6 & 10 sec ? (Ans : 0.18)

Headway → time GAP b/n two successive arrivals.

Sol: $t = \{$ time Gap b/n two successive vehicles in sec $\}$

Av. No. of Vehicles coming at junction in one hr $= 360$ veh.

i.e $\lambda = 360$ veh/hr $= \frac{360}{60}$ veh/min $= \frac{360}{60\times 60}$ veh/sec $= \frac{1}{10}$ veh/sec

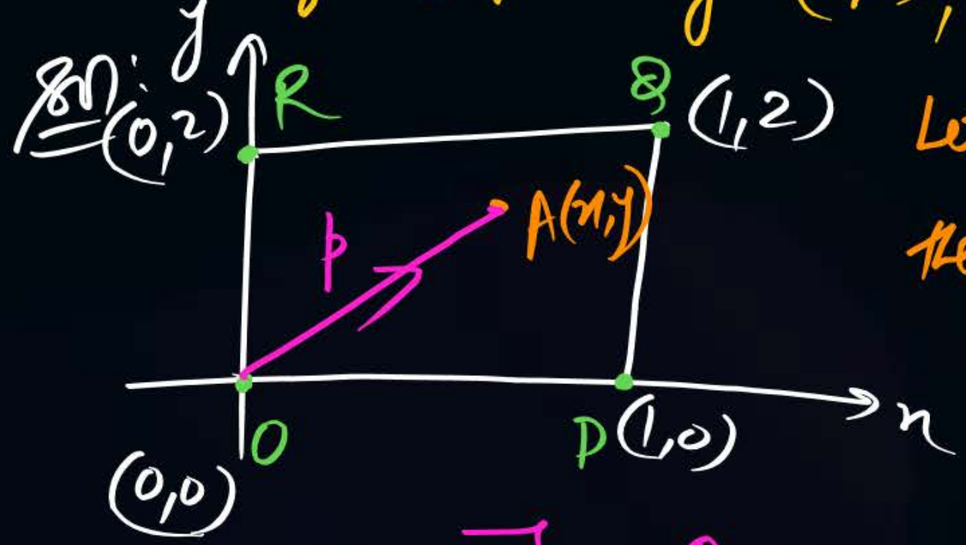
ie Av. time Gap b/n two vehicles in sec $= 10$/sec $= \frac{1}{\mu} \Rightarrow \mu = \frac{1}{10}$ /sec.

$$f(t) = \begin{cases} \mu e^{-\mu t}, & t \geq 0 \\ 0, & t < 0 \end{cases}$$

$$P(6 < x < 10) = \int_6^{10} f(x)\,dx = \int_6^{10} \mu e^{-\mu t}\,dt = \mu\left[\frac{e^{-\mu t}}{-\mu}\right]_6^{10}$$

$$= -\left[e^{-10\mu} - e^{-6\mu}\right] = \frac{1}{e^{6\mu}} - \frac{1}{e^{10\mu}} = \frac{1}{e^{0.6}} - \frac{1}{e^{1}} = 0.12$$

HW8 Let ρ is the length of position vector of any point (x,y) with in the Rectangle formed by $(0,0), (1,0), (1,2), (0,2)$ then $E(\rho^2) = ?$



Let $A(x,y)$ is any Random Point in the Rectangle

then $0 \le x \le 1$ & $0 \le y \le 2$

$$f(x) = \begin{cases} \frac{1}{1-0}, & 0 \le x \le 1 \\ 0, & \text{otw} \end{cases} \quad \& \quad f(y) = \begin{cases} \frac{1}{2-0}, & 0 \le y \le 2 \\ 0, & \text{otw} \end{cases}$$

ATQ, $\vec{OA} = x\hat{i} + y\hat{j} \Rightarrow \rho = |\vec{OA}| = \sqrt{x^2+y^2}$ i.e. $\rho^2 = x^2+y^2$

$\because p^2 = x^2 + y^2 \Rightarrow E(p^2) = E(x^2 + y^2)$

$= E(x^2) + E(y^2)$

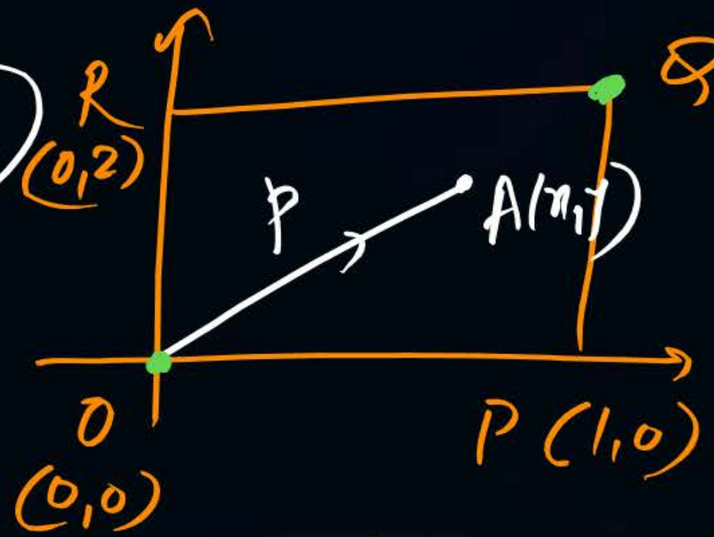
$= \int_{-\infty}^{\infty} x^2 f(x)\,dx + \int_{-\infty}^{\infty} y^2 f(y)\,dy$

$= \int_0^1 x^2 (1)\,dx + \int_0^2 y^2 \left(\frac{1}{2}\right) dy$

$= \left(\frac{x^3}{3}\right)_0^1 + \frac{1}{2}\left(\frac{y^3}{3}\right)_0^2$

$= \left(\frac{1-0}{3}\right) + \frac{1}{2}\left(\frac{8-0}{3}\right) = \frac{10}{6} = \frac{5}{3}$

M-II



Min value of ρ occurs when A is at origin (0,0)

& Max " " " when A is at Q(1,2)

Now in $\triangle OQP \Rightarrow \rho^2 = 1^2 + 2^2 = 5 \Rightarrow \boxed{\rho = \sqrt{5}}$

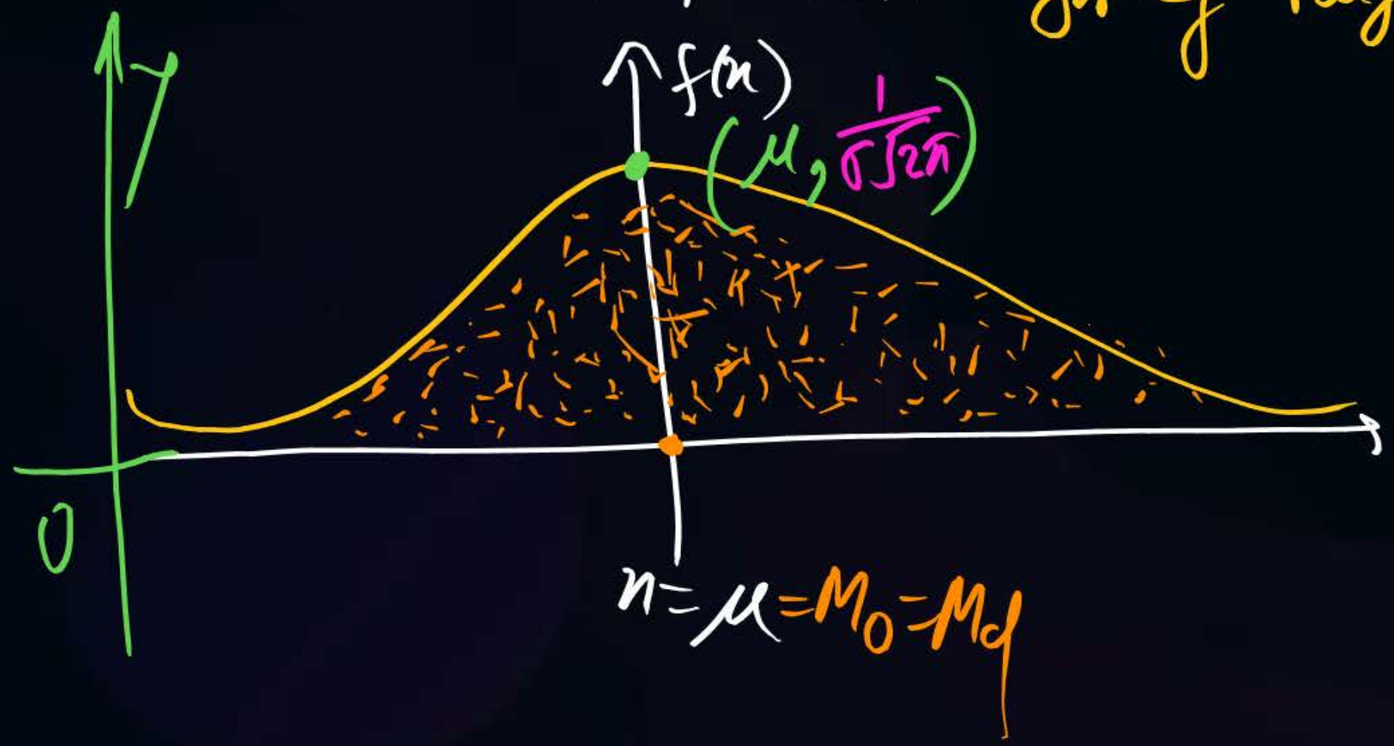
Hence $0 \le \rho \le \sqrt{5}$ & p.d.f of 'ρ' is $f(\rho) = \begin{cases} \frac{1}{\sqrt{5}-0}, & 0 < \rho < \sqrt{5} \\ 0, & \text{otw} \end{cases}$

Hence $E(\rho^2) = \int_{-\infty}^{\infty} \rho^2 f(\rho)\, d\rho = \int_0^{\sqrt{5}} \rho^2 \left(\frac{1}{\sqrt{5}}\right) d\rho = \frac{1}{\sqrt{5}}\left[\frac{\rho^3}{3}\right]_0^{\sqrt{5}} = \frac{5\sqrt{5}}{3\sqrt{5}} = \frac{5}{3}$

NORMAL DIST (GAUSSIAN DIST) / BELL Curve

(≈ 100% they will ask Q from N DIST)

"If Random Variable x has tendency to accumulate about it's Average Value then such types of variables are called N.R.V and the distribution formed is called N. Dist." for eg Height, weight, Age, Intelligence etc



$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} \exp\left\{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2\right\}$$

= NENO func^n

& $f(\mu) = \frac{1}{\sigma\sqrt{2\pi}}$

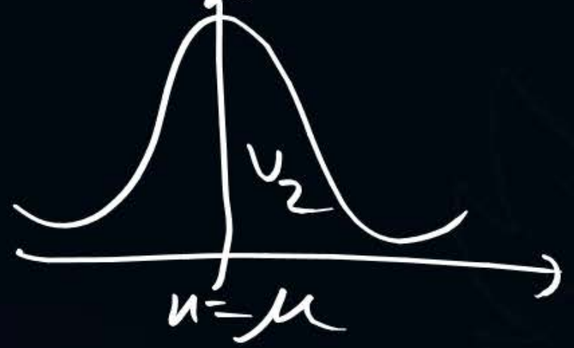
Note ① N. Curve is Symmetrical Curve with Symmetry about $x=\mu$

② N. Curve is Unimodal Curve with highest point occurs at $x=\mu$

$Me = Md = Mo$ & highest Value is $f(\mu) = \frac{1}{\sigma\sqrt{2\pi}}$

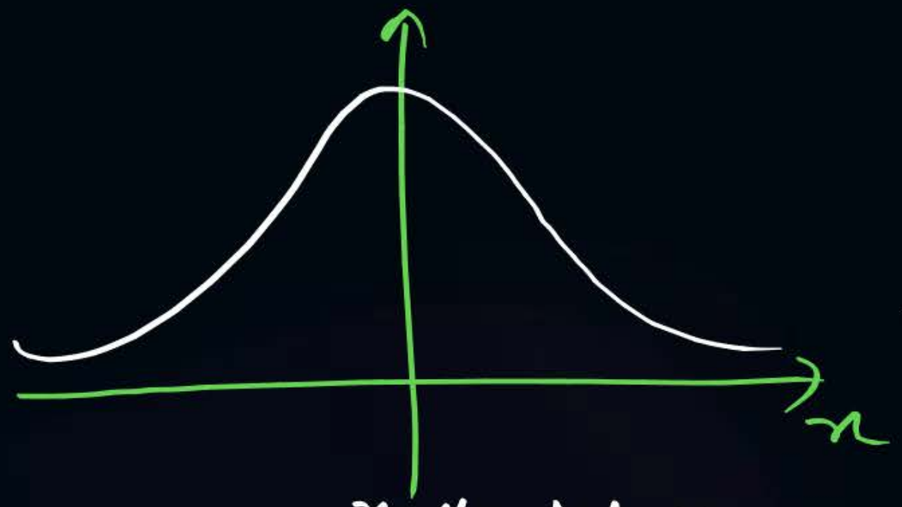
③ N. Curve is Bell Shaped Curve where Variance is Represented by Width of Bell





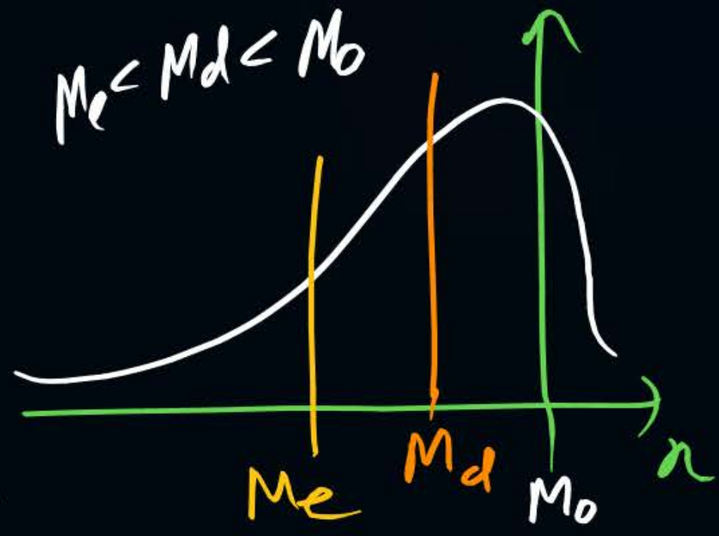
$V_1 > V_2$

(4) Skewness in N. Curve

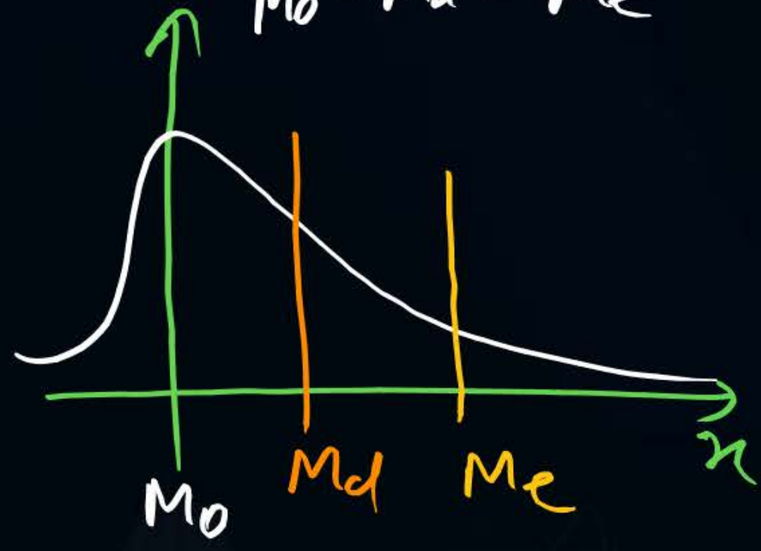


$\bar{x} = \mu = M_d = M_o$

(No Skewness)



(-vely skewed)



(+vely skewed)

Def^n: Let x is C.R.V s.t it's p.d.f is defined as $f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2}$

then x is called N.R.V with parameters μ & σ^2 and it is denoted as $x \sim N\{\mu, \sigma^2\}$

Cross check:- $\int_{-\infty}^{\infty} f(x)dx = \int_{-\infty}^{\infty} \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2} dx = \overset{\text{using Gamma func}^n}{\text{- - - - - - - -}} = 1$ Hence Verified

i.e Total area under $f(x)$ is of 1 unit.

Note (1) $\text{Mean}(x) = E(x) = \int_{-\infty}^{\infty} x f(x)dx = \text{- - - - - - - - } = \mu$

(2) $\text{Var}(x) = E(x^2) - E^2(x) = \text{- - - - - - - - - - - - } = \sigma^2$

(3) $SD = \sigma$

Q. Quest for Revision → If x is a R.V with mean $= \mu$ & S.D $= \sigma$ & $Z = \frac{x-\mu}{\sigma}$

then Find Mean & Var of Z

① $\text{Mean}(Z) = E(Z) = E\left\{\frac{x-\mu}{\sigma}\right\} = \frac{1}{\sigma}\left\{E(x-\mu)\right\} = \frac{1}{\sigma}\left\{E(x) - E(\mu)\right\}$

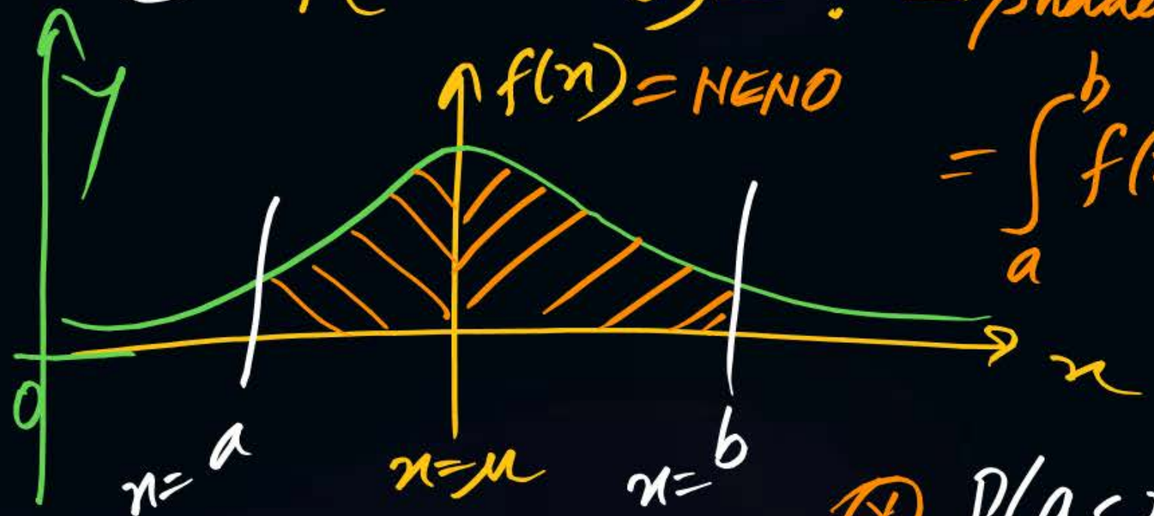
$= \frac{1}{\sigma}\{\mu - \mu\} = 0$

② $\text{Var}(Z) = \text{Var}\left(\frac{x-\mu}{\sigma}\right) = \frac{1}{\sigma^2}\text{Var}(x-\mu) = \frac{1}{\sigma^2}\left[\text{Var}(x) - \text{Var}(\mu)\right]$

$= \frac{1}{\sigma^2}\{\sigma^2 - 0\} = 1$

i.e for $Z = \frac{x-\mu}{\sigma}$, $\mu_Z = 0$, $\sigma_Z = 1$

(*) $P(a<x<b) = ?$ = Shaded Area = ? = Not easy to Calculate

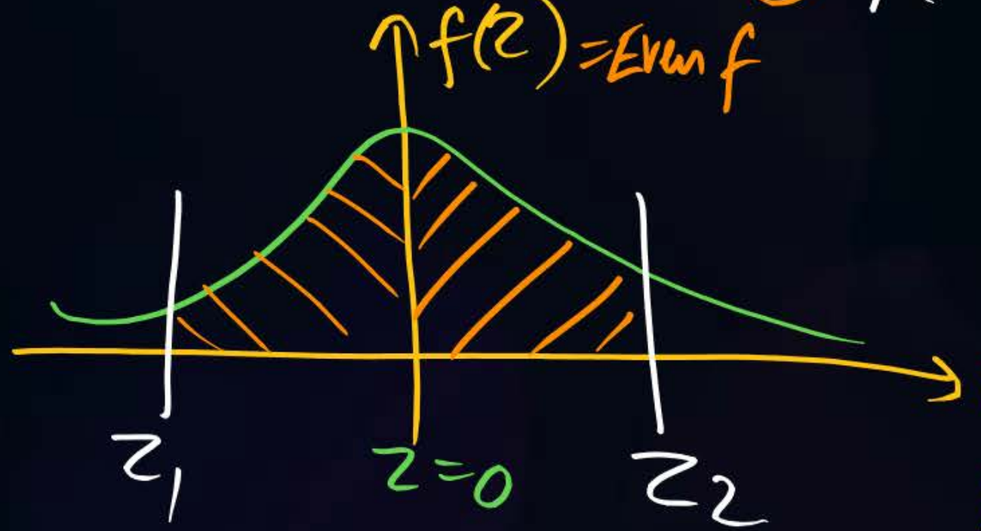


$= \int_a^b f(x)\,dx = ?$ = " " " "

(*) $P(a<x<b) = P(z_1<z<z_2)$

= Area under $f(z)$ b/n z_1 & z_2

= use Normal Table = Ans.



$z = \frac{x-\mu}{\sigma}$

At $x = \mu$, $z = 0$

At $x = a$, $z_1 = \frac{a-\mu}{\sigma}$

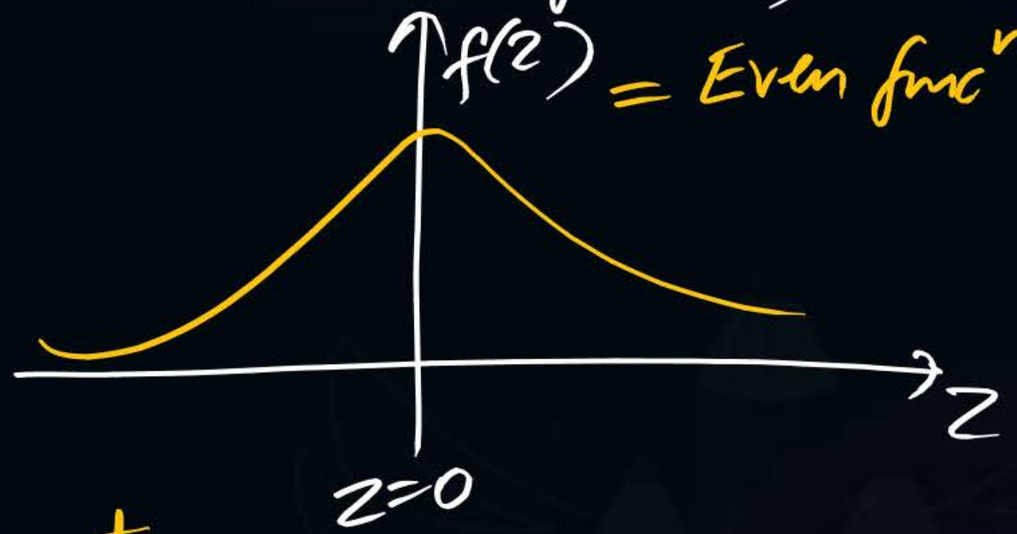
At $x = b$, $z_2 = \frac{b-\mu}{\sigma}$

Standard Normal Variable (Z) → if we take take $\mu = 0, \sigma = 1$ then $\frac{x-\mu}{\sigma} = z$

& z is called S.N. Variable with it's p.d.f is defined as;

$$f(z) = \frac{1}{\sqrt{2\pi}} e^{-\frac{z^2}{2}}$$

is $Z \sim N\{0, 1\}$



$f(z)$ = Even func^n

(1) S.N.V is free from any parameter

Note ① N. Table starts from $Z=0$ and is defined for +ve values of Z

② eg $P(0 \le z \le 2) = ? = \int_0^2 f(z)\,dz = \int_0^2 \frac{1}{\sqrt{2\pi}} e^{-\left(\frac{z^2}{2}\right)} \cdot dz = \cdots = 0.478$ (using Calci)

eg $P(0 \le z \le 1.15) = ? = \int_0^{1.15} f(z)\,dz = \int_0^{1.15} \frac{1}{\sqrt{2\pi}} e^{-\left(\frac{z^2}{2}\right)} \cdot dz = \cdots = 0.3746$ (using Calci)

eg $P(0 \le z \le 1.28) = ? = \int_0^{1.28} f(z)\,dz = \cdots = 0.4$

(*) Flowchart of Solving Questions

Step (1) First find $x = \{$which is Required$\}$

Step (2) Convert x into z using $z = \frac{x - \mu}{\sigma}$

* Step (3) Use Concept of Symmetry

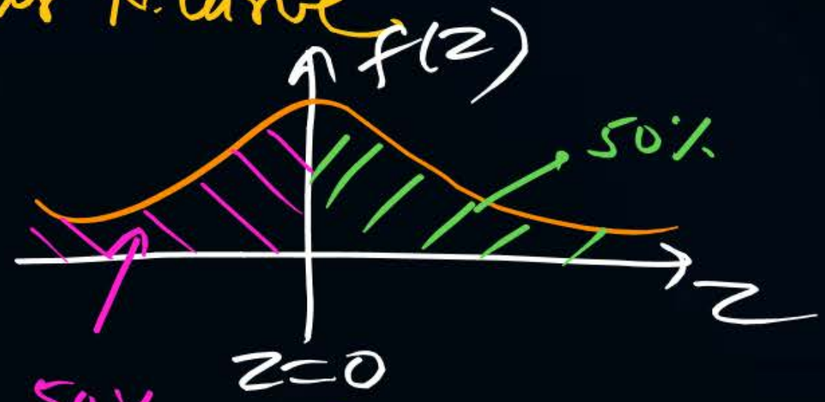
Step (4) Use N. Table

Concept of Symmetry :-

① $P(-\infty < z < \infty) = 1$ i.e Total area under N. Curve

② $P(-\infty < z \leq 0) = \frac{1}{2} = P(0 \leq z < \infty)$

(Left side area) (Right side area)



③ $P(-1.15 \leq z \leq 1.15) = ? = 2P(0 \leq z \leq 1.15)$

$= 2(\text{N. Table})$

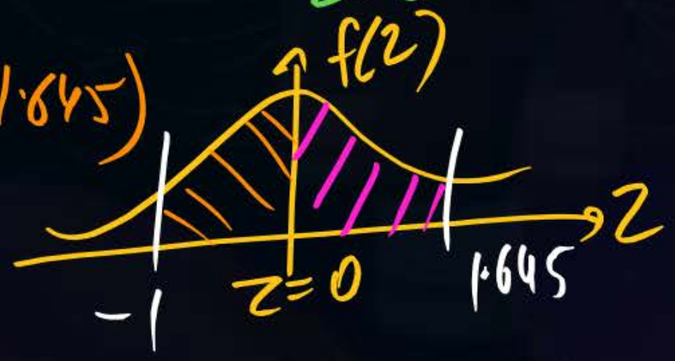
$= 2(0.3746)$



④ $P(-1 \leq z \leq 1.645) = ? = P(-1 \leq z \leq 0) + P(0 \leq z \leq 1.645)$

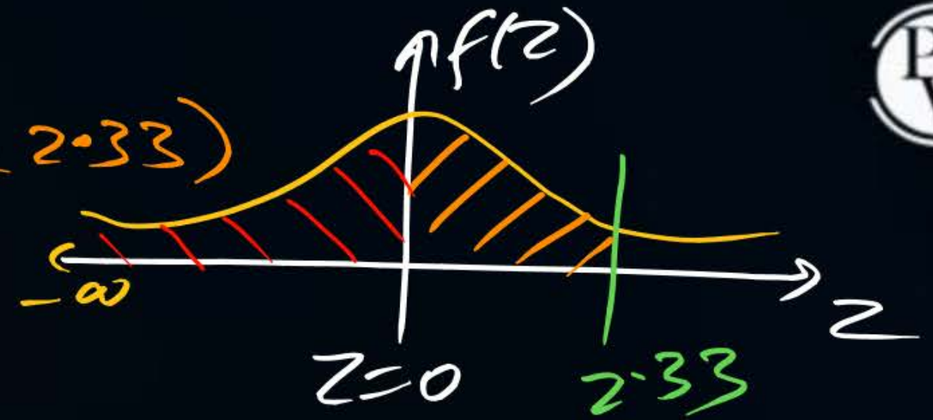
$= P(0 \leq z \leq 1) + \text{N. Table}$

$= 0.3413 + 0.45$



⑤ $P(-\infty < z \leq 2.33) = ? = \frac{1}{2} + P(0 \leq z \leq 2.33)$

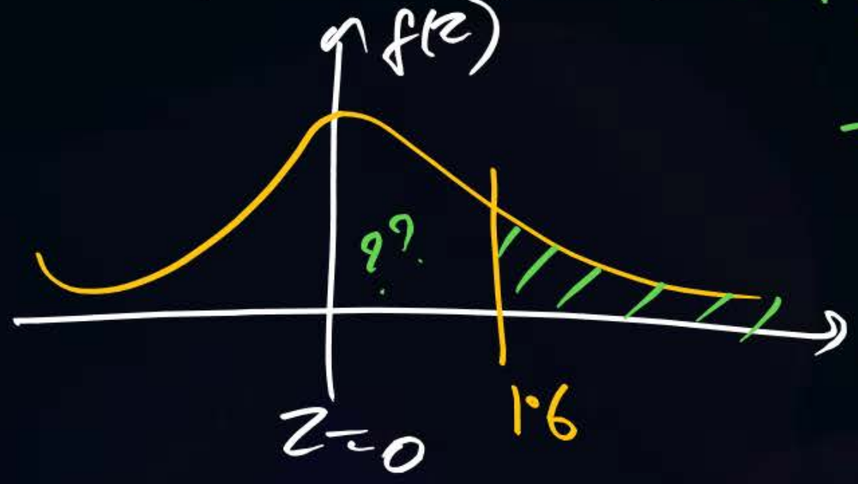
$= \frac{1}{2} + (0.49)$



⑥ $P(z > 1.6) = ?$ = Right side area $- P(0 \leq z \leq 1.6)$

$= \frac{1}{2} - (\text{use N-Table})$

$= \frac{1}{2} - (0.4452) = \underline{\text{Ans}}$



Q: In a test, 2000 students are appearing and Distribution of Marks is assumed to be Normal with Mean 30 marks & S.D 6·25 marks then Find the Number of students getting marks b/n 20 & 40.

It is given that area under the Normal Curve b/n $Z=0$ & $Z=1.6$ is 0·4452

Sol: $N=2000$ students, for single student : $x=\{\text{Marks obtained}\}$

$\mu=30, \sigma=6.25$, $Z=\frac{x-\mu}{\sigma}$

$Z_1=\frac{20-30}{6.25}=-1.6$

$Z_2=\frac{40-30}{6.25}=1.6$

$P(20<x<40)=P(-1.6<Z<1.6)=2P(0<Z<1.6)=2(0.4452)=0.8904$

$$\text{i.e. } P(20 < x < 40) = \frac{0.8904}{1} = \frac{890.4}{1000} = \frac{1780.8}{2000} \approx \frac{1781}{2000}$$

(a) 890

(b) 1780

(c) 1781

(d) 891

i.e. 1781 students are getting marks b/w 20 & 40.

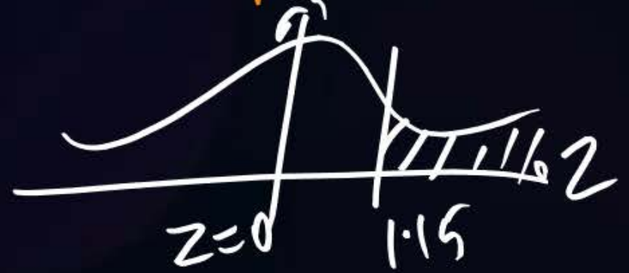
Note:- large number of students are justifying the def^n of N. Dist ($\because$ Av Marks = 30)

Q. In a school of 1000 students, mean height of student is 68.22 inches and variance 10.8 (inches)^2 then find the number of students to be over 6 feet tall if it is given that we have confidence limit of 37.46% from $Z=0$ to 1.15 ?

Sol: $N = 1000$ student, For single student: $x = \{$ Height of student $\}$

$\mu = 68.22$ inches, $\sigma = \sqrt{10.8}$, $Z = \dfrac{x-\mu}{\sigma} = \dfrac{72-68.22}{\sqrt{10.8}} = 1.15$

$P(x > 72) = P(Z > 1.15) = \dfrac{1}{2} - P(0 \le Z \le 1.15) = 0.5 - 0.3746 = 0.1254 = \dfrac{125}{1000}$



i.e No. of students to be over 6 feet tall = 125

$\therefore P(x > 72) = \frac{0.1254}{1} = \frac{125.4}{1000} \simeq \frac{125}{1000}$

$\therefore$ No. of students to be over 6 feet tall $= 125$ Ans

Note:-

	Given parameters	Dist
(1)	n & p	Binomial
(2)	λ	Poisson
(3)	μ	Exp.
(4)	$x \in (a, b)$	Uniform.
(5)	μ & σ^2	Normal.

THANK - YOU